



TECHMINDS ACADEMY

Node.js & Backend Development

Server-Side Development Curriculum · Intermediate to Job-Ready

A structured 15-week programme for students who have completed the TechMinds JavaScript curriculum. From running your first Node.js script to deploying a production-ready, authenticated, tested REST API.

15 Weeks	5 Stages	70+ Lessons	9 Projects	1 Prerequisite
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Node.js Core · Express.js · MongoDB & Mongoose · PostgreSQL & Prisma · JWT Auth · Security · File Uploads · Jest & Supertest · Deployment

LEARNING PATHWAY

Programme Overview

Stage	Focus	Duration	Level	Prerequisite
Stage 1	Node.js Foundations — modules, fs, HTTP server	3 weeks	Beginner	JavaScript
Stage 2	Express.js — REST APIs and middleware	3 weeks	Beginner	Stage 1
Stage 3	Databases — MongoDB & PostgreSQL	3 weeks	Intermediate	Stage 2
Stage 4	Authentication & Security	3 weeks	Intermediate	Stage 3
Stage 5	Advanced Features, Testing & Capstone	3 weeks	Advanced	Stage 4

CURRICULUM RATIONALE

Key Decisions Explained

Why Node.js for backend?	Node.js lets students use the same language — JavaScript — on both frontend and backend. This dramatically reduces the learning curve compared to switching to Python or PHP. It is also widely used in Nigerian and global tech companies.
Why cover both MongoDB and PostgreSQL?	Real jobs require flexibility. MongoDB is common in startups and fast-moving products. PostgreSQL is the standard for data-sensitive applications. Knowing both makes the student employable across a much wider range of roles.
Why include security as a full stage?	Security is the most overlooked topic in beginner backend courses — and the most costly to skip. Three weeks ensures the student understands authentication, authorisation, and hardening before they deploy anything to production.
What comes after this course?	Node.js Backend feeds into the TechMinds Full-Stack course — combining React on the frontend with Node.js on the backend. Students can also specialise further with the TechMinds DevOps and Cloud Infrastructure course.

Stage 01 · Beginner

3 Weeks · Weeks 1 – 3

Node.js Foundations

Running JavaScript on the server and understanding the Node environment

This stage assumes the student has completed the TechMinds JavaScript curriculum. Node.js is introduced from the ground up — what it is, how it differs from browser JavaScript, and how to use it to run server-side code. By the end of Week 3 the student will be building basic HTTP servers and working with the filesystem.

Week 1 Introduction to Node.js

5 lessons · 6 hrs

01 What is Node.js? How it runs JavaScript outside the browser

02 Node.js vs browser JS — what's different and what's missing

03 Installing Node.js and using the Node REPL

04 Running JavaScript files with node filename.js

05 The global object, process, and __dirname

06 npm — Node Package Manager and the node_modules folder

PROJECT

A CLI tool that accepts a name argument and prints a personalised greeting

Week 2 Modules & the File System

5 lessons · 6 hrs

01 CommonJS modules — require() and module.exports

02 ES Modules in Node.js — import and export with type:module

03 Built-in modules: path, os, and url

04 The fs module — reading and writing files synchronously

05 Async file operations — fs.promises and async/await

06 Working with JSON files — read, parse, modify, and write

PROJECT

A simple notes CLI — create, list, read, and delete notes stored in a JSON file

Week 3 HTTP & Your First Server

5 lessons · 7 hrs

01 How the web works — requests, responses, and HTTP methods

02 The http module — creating a basic server with Node.js

03 Routing requests manually — if/else on req.url

04 Sending HTML, JSON, and plain text responses

05 HTTP status codes — 200, 201, 400, 401, 404, 500

06 Reading request body data from POST requests

PROJECT

A raw Node.js HTTP server with 3 routes: GET /, GET /about, POST /data

Stage 02 · Beginner

3 Weeks · Weeks 4 – 6

Express.js — Building REST APIs

The industry-standard Node.js framework for building web servers and APIs

Express.js is the most widely used Node.js framework in the world. This stage replaces the raw http module with Express and introduces the patterns every backend developer must know — routing, middleware, request/response handling, and building a clean REST API.

Week 4

Express.js Fundamentals

5 lessons · 6 hrs

01 What is Express.js and why use it over raw Node http?

02 Installing Express and setting up a project with npm

03 Defining routes: GET, POST, PUT, PATCH, DELETE

04 Route parameters — /users/:id

05 Query strings — /products?category=shoes&sort=price

06 Sending JSON responses with res.json()

PROJECT

A products REST API — GET all, GET one, POST, PUT, DELETE using in-memory data

Week 5

Middleware

5 lessons · 6 hrs

01 What is middleware? The request-response cycle in Express

02 app.use() — applying middleware globally

03 Built-in middleware: express.json() and express.static()

04 Writing custom middleware — logging, timing, validation

05 Route-level middleware — applying only to specific routes

06 Error-handling middleware — the 4-argument pattern

PROJECT

Add request logging, JSON body parsing, and a custom error handler to the products API

Week 6

REST API Design & Express Router

5 lessons · 7 hrs

01 REST principles — resources, statelessness, uniform interface

02 Designing clean API endpoints and URL structures

03 Express Router — splitting routes into separate files

04 Organising a project: routes/, controllers/, middleware/

05 Input validation with express-validator

06 Testing API endpoints with Thunder Client or Postman

PROJECT

A fully structured Express REST API for a blog — posts and comments, split into router files

Stage 03 · Intermediate

3 Weeks · Weeks 7 – 9

Databases — MongoDB & PostgreSQL

Persisting data with both NoSQL and relational databases

A backend without a database stores nothing permanently. This stage covers two databases — MongoDB with Mongoose for NoSQL, and PostgreSQL with Prisma for relational data. Students learn to model data, write queries, and connect their Express APIs to a real database for full persistence.

Week 7

MongoDB & Mongoose

5 lessons · 7 hrs

01 What is MongoDB? Documents, collections, and the NoSQL model

02 Setting up MongoDB Atlas — free cloud database

03 Connecting Express to MongoDB with Mongoose

04 Defining schemas and models with Mongoose

05 CRUD operations: create, find, findById, updateOne, deleteOne

06 Population — referencing documents across collections

PROJECT

Connect the Week 6 blog API to MongoDB — posts and comments fully persisted

Week 8

PostgreSQL & Prisma

5 lessons · 7 hrs

01 What is PostgreSQL? Tables, rows, and relational data

02 Setting up a PostgreSQL database locally and on Railway

03 Prisma ORM — schema-first database access for Node.js

04 Defining models in schema.prisma — fields, types, relations

05 Prisma Client — create, findMany, update, delete

06 Relations: one-to-many and many-to-many with Prisma

PROJECT

A task manager API with PostgreSQL and Prisma — users, tasks, and categories with relations

Week 9

Database Best Practices

5 lessons · 6 hrs

01 Environment variables — storing DB credentials safely with dotenv

02 Database indexing — speeding up common queries

03 Pagination — limit and skip / cursor-based pagination

04 Filtering and sorting API results from the database

05 Transactions — all-or-nothing database operations

06 Seeding a database — populating test data automatically

PROJECT

Add pagination, filtering, sorting, and dotenv config to the task manager API

Stage 04 · Intermediate

3 Weeks · Weeks 10 – 12

Authentication & Security

Protecting APIs with JWT authentication and security best practices

Security is non-negotiable in backend development. This stage teaches authentication from scratch — password hashing, JWT tokens, protected routes, and refresh token rotation. Security best practices including rate limiting, CORS, helmet, and input sanitisation are covered in depth.

Week 10

Authentication — Passwords & JWT

5 lessons · 7 hrs

01 Why never store plain text passwords — hashing with bcrypt

02 User registration — hash password and save to database

03 User login — compare password hash and verify identity

04 What is JWT? Header, payload, and signature explained

05 Signing and verifying JWTs with jsonwebtoken

06 Sending the token to the client and storing it safely

PROJECT

A full auth system — register, login, and receive a JWT; built on the task manager API

Week 11

Protected Routes & Refresh Tokens

5 lessons · 7 hrs

01 Auth middleware — verifying the JWT on every protected request

02 Attaching the user to req.user after token verification

03 Role-based access control — admin vs standard user

04 Access tokens vs refresh tokens — why both are needed

05 Implementing refresh token rotation

06 Logout — invalidating tokens and clearing cookies

PROJECT

Add protected routes, role-based access, and refresh token support to the auth system

Week 12

API Security Best Practices

5 lessons · 6 hrs

01 Helmet.js — setting secure HTTP headers automatically

02 CORS — controlling which origins can access the API

03 Rate limiting with express-rate-limit — preventing abuse

04 Input sanitisation — preventing NoSQL and XSS injection

05 Hiding sensitive data — removing password fields from responses

06 Security checklist — what to verify before going live

PROJECT

Harden the task manager API — add Helmet, CORS config, rate limiting, and sanitisation

Stage 05 · Advanced

3 Weeks · Weeks 13 – 15

Advanced Node.js, Testing & Capstone

Production-grade patterns, automated testing, and deploying a real backend

The final stage covers the skills that make a backend developer production-ready. File uploads, email sending, background jobs, and WebSockets are introduced. API testing with Jest and Supertest ensures the student can verify their code. The course closes with a capstone — a complete, deployed, documented REST API.

Week 13

Advanced Features

5 lessons · 8 hrs

01 File uploads with Multer — images and documents

02 Storing files in Cloudinary — cloud image management

03 Sending emails with Nodemailer and Resend

04 Background jobs and queues with Bull and Redis

05 WebSockets with Socket.io — real-time communication

06 Caching API responses with Redis for performance

PROJECT

Add image upload (Multer + Cloudinary) and welcome email (Nodemailer) to the auth system

Week 14

Testing Node.js APIs

5 lessons · 7 hrs

01 Why test a backend? Unit vs integration vs end-to-end

02 Setting up Jest for Node.js testing

03 Supertest — making HTTP requests in tests without a live server

04 Testing GET, POST, PUT, DELETE endpoints

05 Mocking the database — testing without a real DB connection

06 Test coverage — measuring how much code is tested

PROJECT

Write a full test suite for the task manager API — all CRUD routes covered with Jest and Supertest

Week 15

Capstone Project

6 sessions · 12 hrs

01 Planning the API — resources, endpoints, auth, database

02 Project setup — Express, Prisma or Mongoose, folder structure

03 Building core routes — CRUD with full validation

04 Adding authentication — register, login, protected routes

05 Writing tests — at least 80% route coverage

06 Deployment — Railway or Render, environment variables, live URL

PROJECT

A complete production-ready REST API — student chooses: food delivery, e-commerce, or event booking backend; deployed and documented

ASSESSMENT & COMPLETION

How Students Are Evaluated

Component	Description	Weight
Weekly Projects (×9)	API or backend feature built from scratch each week	40%
Stage Assessments (×5)	Short quiz and API code review at end of each stage	30%
Capstone API	Full production REST API — deployed, tested, and documented	30%

LEARNING OUTCOMES

What Students Can Do on Completion

On completing this programme, the student will be able to:

- Run and write JavaScript on the server using Node.js
- Build clean, well-structured REST APIs with Express.js
- Design and query both MongoDB (Mongoose) and PostgreSQL (Prisma) databases
- Implement full JWT-based authentication with refresh token rotation
- Apply role-based access control to protect API routes
- Harden an API with Helmet, CORS, rate limiting, and input sanitisation
- Handle file uploads and integrate cloud storage with Cloudinary
- Send transactional emails and implement background job queues
- Write automated API tests using Jest and Supertest
- Deploy a production Node.js API to Railway or Render with environment variables

TOOLS & RESOURCES

Everything the Student Needs

Tool	Purpose	Cost
Node.js + npm	JavaScript runtime and package manager	Free
Express.js	Web framework for building REST APIs	Free
MongoDB Atlas	Cloud-hosted MongoDB database	Free tier
PostgreSQL + Railway	Relational database with free cloud hosting	Free tier
Prisma ORM	Type-safe database access for PostgreSQL	Free

Thunder Client / Postman	Test and document API endpoints	Free
Jest + Supertest	Automated testing for Node.js REST APIs	Free
Railway / Render	Deploy and host the capstone API with a live URL	Free tier

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